P A L E T T E

Pixel & Pages

PROJECT DETAILS

- **Name:** Kaviya S
- **Roll No:** E25AI118
- **Department:** B.Sc Computer Science With AI

REQUIREMENT ANALYSIS REPORT

Project: E-Commerce Bookstore Database Management System

1. INTRODUCTION

As our e-commerce bookstore continues to experience rapid growth, the existing methods for managing daily operations are no longer sufficient. To scale effectively, the company requires a robust, centralized Database Management System (DBMS) capable of handling high volumes of data across multiple business domains.
This document serves as the formal Requirement Analysis for the proposed e-commerce database. It outlines the specific data requirements, business constraints, and operational logic needed to support everything from user registration and inventory tracking to complex payment processing and business analytics. The objective of this phase is to clearly define what data needs to be stored and managed before we move into the architectural design and modeling phases.

2. SYSTEM OVERVIEW & SCOPE

The proposed database will serve as the core backend for an e-commerce bookstore platform. The system must seamlessly handle a hybrid catalog consisting of both digital assets (e-books) and physical goods (paperbacks/hardcovers). The database must manage high-volume transactional data, ensure referential integrity, and support analytical reporting for management.

3. USER ROLES

The database must support operations for two primary types of users:

- **Customers:** Can browse products, manage their profiles, place orders, track shipments, and leave reviews.
- **Administrators:** Can add/remove books, update inventory levels, process shipments, and run financial reports.

4. FUNCTIONAL REQUIREMENTS

4.1 Customer Registration and Profile Management

Description: The system must capture and store user identities, securely manage authentication, and handle multiple delivery addresses.

- **Key Data Points Needed:** User ID, First Name, Last Name, Email Address, Hashed Password, Phone Number, Account Creation Date.
    - Address points: Street, City, State, Pincode, Address Type (Billing/Shipping).
- **Business Rules & Constraints:**
    - Email Address must be UNIQUE and NOT NULL.
    - Passwords must never be stored in plain text.
    - A single customer can have multiple saved addresses (1-to-Many relationship).

4.2 Product and Category Management

Description: The database acts as a master catalog for all books. It must handle complex relationships, such as a book having multiple authors or belonging to multiple genres.

- **Key Data Points Needed:** Book ID, ISBN, Title, Author(s), Publisher, Publication Date, Format (Physical/E-Book), File Size (for E-books), Page Count, Unit Price, Description.
    - Category points: Category ID, Category Name.
- **Business Rules & Constraints:**
    - ISBN must be UNIQUE.
    - Unit Price must be greater than zero.
    - Books and Categories share a Many-to-Many relationship (e.g., A book can be both "Technology" and "Cyber Security").

4.3 Inventory Tracking

Description: The system must monitor stock levels in real-time to prevent users from ordering out-of-stock physical items.

- **Key Data Points Needed:** Inventory ID, Book ID, Current Stock Quantity, Warehouse Location, Reorder Threshold.
- **Business Rules & Constraints:**
    - Current Stock Quantity cannot drop below zero.
    - If a book format is "E-Book", inventory tracking is bypassed (stock is essentially infinite).
    - The system should flag an alert when Current Stock <= Reorder Threshold.

4.4 Order Placement and Order History

Description: The core transactional module. It must lock in the price of a book at the exact time of purchase and support multiple items per order.

- **Key Data Points Needed:**
    - **Order Master:** Order ID, User ID, Order Date/Time, Total Amount, Overall Status (Pending, Confirmed, Cancelled).

    - **Order Line Items:** Order Item ID, Order ID, Book ID, Quantity, Price at Purchase.
- **Business Rules & Constraints:**
    - An order must contain at least one item.
    - Price at Purchase must be recorded separately from the book's current catalog price, so past order totals don't change if the book's price goes up later.

4.5 Payment Processing and Invoicing

Description: Financial tracking for the system. It records the payment gateway response and generates invoice records.

- **Key Data Points Needed:** Payment ID, Order ID, Payment Method (Card, UPI, Wallet), Transaction Reference ID, Payment Date, Payment Status (Success, Failed, Refunded), Tax Amount, Final Billed Amount.
- **Business Rules & Constraints:**
    - Payment ID is heavily linked to Order ID (1-to-1 relationship).
    - An invoice is only generated if the Payment Status is 'Success'.

4.6 Shipment Tracking

Description: Logistics module for physical book orders. It maps the handover of goods to third-party couriers.

- **Key Data Points Needed:** Shipment ID, Order ID, Courier Partner Name, Tracking Number, Dispatch Date, Estimated Delivery Date, Current Shipment Status.
- **Business Rules & Constraints:**
    - Digital-only orders (e-books) will not generate a shipment record.
    - Tracking Number should be logged as a UNIQUE string to prevent duplicate tracking entries.

4.7 Product Reviews and Ratings

Description: User-generated content linking customers to the products they have purchased.

- **Key Data Points Needed:** Review ID, Book ID, User ID, Rating Value, Review Text, Timestamp, Approval Status.
- **Business Rules & Constraints:**
    - Rating Value is restricted to integers between 1 and 5.
    - A user can only leave one review per specific Book ID.
    - **Constraint:** The database should verify if the User ID has a "Success" order containing the Book ID before allowing a review (Verified Buyer check).

4.8 Sales Reporting and Business Analytics

Description: The data structure must be optimized to allow admins to extract business intelligence efficiently using SQL aggregation functions (SUM, COUNT, GROUP BY).

- **Query Requirements (To be supported by the schema):**
    - Calculate total revenue generated per day/week/month.
    - Identify the top 10 most frequently purchased books.
    - Calculate the conversion rate of shopping carts to successful checkouts.
    - Generate a list of physical books currently sitting below the Reorder Threshold.

5. NON-FUNCTIONAL REQUIREMENTS & DATA INTEGRITY

To ensure a reliable database architecture, the design will adhere to strict DBMS principles:

- **Normalization:** The schema will be normalized to the Third Normal Form (3NF) to eliminate data redundancy (e.g., storing the customer's full address in a separate table rather than duplicating it on every single order row).
- **Referential Integrity:** Strict Foreign Key constraints will ensure that an Order cannot exist without a valid User, and an Order Item cannot exist without a valid Book.
- **ACID Compliance:** All checkout and payment data insertions will be wrapped in transaction blocks to guarantee Atomicity (either the whole order succeeds, or none of it does) and Consistency.

6. CONCLUSION

This requirement analysis successfully captures the critical data needs of the expanding e-commerce bookstore. By breaking down the daily operations into distinct functional modules ranging from customer profile management to backend sales reporting we have established a clear blueprint for the database architecture. The constraints and business rules identified here will ensure data integrity, prevent anomalies, and guarantee that the system can handle both digital and physical product lifecycles.